

Danesh Morales Hashemi

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Research Positions

- 2025 – Present **Graduate Research Assistant**, Institute for Quantum Computing
Advisors: [Graeme Smith](#) and [Ningping Cao](#)
- 2024 – 2025 **Undergraduate Research Assistant**, Institute for Quantum Computing
Advisor: [Joseph Emerson](#)
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*
- Fall 2023 **Undergraduate Research Assistant**, University of Waterloo
Advisor: [Roberto Guglielmi](#)
Topic: *Controllability of Fokker-Planck Equation*

Education

- 2025 – Present **MMath in Applied Mathematics**, University of Waterloo
Quantum Information Specialization
- 2020 – 2025 **BMath in Mathematical Physics**, University of Waterloo
Co-op Program • Pure Mathematics Minor • Graduated with [Dean's Honours](#)

Publications

1. **What Must a Quantum-Memory Decoder Know About Temporally Correlated Noise?**
Danesh Morales-Hashemi, Graeme Smith, and Ningping Cao.
Preprint, 2026 • [arXiv:2610.03545](#) [quant-ph]

Conferences and Talks

1. **Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation**
APS March Meeting 2025
Session on Scalable Verification and Validation (QCVV)
Anaheim Convention Center • March 2025

Teaching Experience

Spring 2025	Teaching Assistant , AMATH 250 – Introduction to Differential Equations
Fall 2023	Instructional Support Assistant , MATH 115 – Linear Algebra for Engineering
Winter 2022	Instructional Support Assistant , CS 115 – Introduction to Computer Science

Projects

- 1. Introduction to Cycle Benchmarking for Quantum Error Diagnostics**
QIC 820 Final Project, Institute for Quantum Computing • December 2025
- 2. Predicting MLB Playoffs Using Quantum Machine Learning**
QIC 890 Final Project, Institute for Quantum Computing • December 2025
 [GitHub repository](#)
- 3. Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation**
Undergraduate Thesis, University of Waterloo • April 2025
- 4. Introduction to Superoperator Representations and Characterization of Completely Positive Maps**
AMATH 900 Final Project, University of Waterloo • August 2024

Awards and Scholarships

Institute for Quantum Computing Undergraduate Research Award 2024	\$16,000 CAD
University of Waterloo President's Scholarship 2020	\$2,500 CAD
Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama	\$250,000 USD

Relevant Coursework

(with links to syllabus)

Quantum Physics	Quantum Theory 1/2/3	Representation Theory
Quantum Information Processing	Quantum Machine Learning	Groups and Ring Theory
Quantum Devices	Theory of Quantum Information	Galois and Field Theory